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Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: 5H Farms #2

Physical address of dairy:

Number and Street	City	County	Zip Code
2163 Rahilly RD	Merced	Merced	95341

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 05/11/1941

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X065-X130-X065-XXXX

B. OPERATORS

Vyfhuizen, Case

Operator name: Vyfhuizen, Case

Telephone no.: Landline: ; Cellular: (209) 631-1887

Mailing Address Number and Street	City	State	Zip Code
5851 S Highway 59	Merced	CA	95341

This operator is responsible for paying permit fees.

C. OWNERS

Vyfhuizen, Case

Legal owner name: Vyfhuizen, Case

Telephone no.: Landline: ; Cellular: (209) 631-1887

Mailing Address Number and Street	City	State	Zip Code
5851 S Highway 59	Merced	CA	95341

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	490	400	0	0
Number under roof	0	0	0	0	0	0
Maximum number	0	0	500	400	0	0
Average number	0	0	490	400	0	0
Avg live weight (lbs)	0	0	900	650		

Predominant milk cow breed: Holstein

Average milk production: 1 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 8,682.80 *tons per reporting period*

Total nitrogen from manure: 84,461.00 *lbs per reporting period*

After ammonia losses (30% loss applied): 59,122.70 *lbs per reporting period*

Total phosphorus from manure: 14,293.40 *lbs per reporting period*

Total potassium from manure: 1.00 *lbs per reporting period*

Total salt from manure: 0.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: _____ *gallons*

Total nitrogen generated: _____ *lbs*

Total phosphorus generated: _____ *lbs*

Total potassium generated: _____ *lbs*

Total salt generated: _____ *lbs*

	0	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	0	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Irrigation Wells	Ground water
IW 13	Ground water
IW 14	Ground water
IW 15	Ground water

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E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
07/07/2023	Solid commercial fertilizer *UN 32*	4.40 *ton*	As-is	0.1	32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	2,816.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	2,816.00	0.00	0.00	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

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APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
23	12	12	2	manure	X065-X130-X065-XXXX
24	17	16	2	manure	X065-X130-X065-XXXX
25	36	34	2	manure	X065-X130-X065-XXXX
26	33	30	2	manure	X065-X130-X065-XXXX
27	38	35	2	manure	X065-X130-X065-XXXX
28	78	78	1	manure	X065-X130-X065-XXXX
Totals for areas that were used for application	214	205	11		
Totals for areas that were not used for application					
Land application area totals	214	205	11		

B. CROPS AND HARVESTS

23

Field name: 23

10/23/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 12 Plant date: 10/23/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	155.20 *ton*	As-is		60.6	4,000.00	1,100.00	6,800.00		10.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.93	103.47	28.45	175.89	1,080.30

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23

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 12 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/12/2023	345.70 *ton*	As-is		62.5	5,400.00	900.00	4,500.00		7.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	32.00	256.00	48.00	211.20	0.00
Total actual harvest content	28.81	311.13	51.86	259.28	1,663.68

24

Field name: 24

10/23/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 16 Plant date: 10/23/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	248.20 *ton*	As-is		59.4	4,000.00	900.00	6,200.00		9.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.51	124.10	27.92	192.36	1,196.63

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 16 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/13/2023	327.00 *ton*	As-is		57.7	6,000.00	800.00	4,500.00		7.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	32.00	256.00	48.00	211.20	0.00
Total actual harvest content	20.44	245.25	32.70	183.94	1,244.89

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25

Field name: 25

10/23/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 34 Plant date: 10/23/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	541.10 *ton*	As-is		56.2	5,200.00	1,200.00	8,800.00		11.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.91	165.51	38.20	280.10	1,547.48

06/10/2023: Corn, silage

Crop: Corn, silage Acres planted: 34 Plant date: 06/10/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/23/2023	768.10 *ton*	As-is		57.5	3,800.00	500.00	4,700.00		5.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	32.00	256.00	48.00	211.20	0.00
Total actual harvest content	22.59	171.69	22.59	212.36	1,113.75

26

Field name: 26

10/23/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 30 Plant date: 10/23/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	426.40 *ton*	As-is		65.6	4,000.00	1,000.00	7,000.00		10.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.21	113.71	28.43	198.99	1,026.77

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26

06/03/2023: Corn, silage

Crop: Corn, silage Acres planted: 30 Plant date: 06/03/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/12/2023	658.30 *ton*	As-is		61.1	5,200.00	800.00	4,300.00		6.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	32.00	256.00	48.00	211.20	0.00
Total actual harvest content	21.94	228.21	35.11	188.71	1,160.89

27

Field name: 27

10/24/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 35 Plant date: 10/24/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	460.50 *ton*	As-is		65.1	4,100.00	1,000.00	6,300.00		10.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	13.16	107.89	26.31	165.78	1,001.02

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 35 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/03/2023	961.50 *ton*	As-is		62.6	4,100.00	500.00	4,000.00		5.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	64.00	12.00	52.80	0.00
Total actual harvest content	27.47	225.27	27.47	219.77	1,109.63

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28

Field name: 28

10/12/2022: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 78 Plant date: 10/12/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/02/2023	1,466.05 *ton*	As-is		54.9	17,600.00	1,800.00	14,400.00		11.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	18.80	661.60	67.66	541.31	1,898.80

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NUTRIENT BUDGET

A. LAND APPLICATIONS

23 - 10/23/2022: Wheat, silage, soft dough

Field name: 23

Crop: Wheat, silage, soft dough — Plant date: 10/23/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/10/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	120.96	23.22	199.26	0.00	40.00 *ton*
Application event totals		120.96	23.22	199.26	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/15/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	5.74	0.00	0.00	389.21	699,600.00 *gal*
Irrigation Wells	Ground water	7.09	0.00	0.00	478.62	871,200.00 *gal*
Application event totals		12.83	0.00	0.00	867.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	4.96	0.00	0.00	336.14	604,200.00 *gal*
Application event totals		4.96	0.00	0.00	336.14	

23 - 06/01/2023: Corn, silage

Field name: 23

Crop: Corn, silage — Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/08/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	184.08	42.48	295.47	0.00	80.00 *ton*
Application event totals		184.08	42.48	295.47	0.00	

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23 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	8.35	0.00	0.00	566.12	1,017,600.00 *gal*
Irrigation Wells	Ground water	10.31	0.00	0.00	696.17	1,267,200.00 *gal*
Application event totals		18.66	0.00	0.00	1,262.30	

06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	6.78	0.00	0.00	459.98	826,800.00 *gal*
Irrigation Wells	Ground water	8.38	0.00	0.00	565.64	1,029,600.00 *gal*
Application event totals		15.16	0.00	0.00	1,025.62	

07/07/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	25.60	0.00	0.00	0.00	
Application event totals		25.60	0.00	0.00	0.00	

07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	8.09	0.00	0.00	548.43	985,800.00 *gal*
Irrigation Wells	Ground water	9.99	0.00	0.00	674.42	1,227,600.00 *gal*
Application event totals		18.08	0.00	0.00	1,222.85	

07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	7.83	0.00	0.00	530.74	954,000.00 *gal*
Irrigation Wells	Ground water	9.67	0.00	0.00	652.66	1,188,000.00 *gal*
Application event totals		17.49	0.00	0.00	1,183.40	

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23 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	5.74	0.00	0.00	389.21	699,600.00 *gal*
Irrigation Wells	Ground water	7.09	0.00	0.00	478.62	871,200.00 *gal*
Application event totals		12.83	0.00	0.00	867.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	6.00	0.00	0.00	406.90	731,400.00 *gal*
Irrigation Wells	Ground water	7.41	0.00	0.00	500.37	910,800.00 *gal*
Application event totals		13.41	0.00	0.00	907.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	5.22	0.00	0.00	353.83	636,000.00 *gal*
Irrigation Wells	Ground water	6.44	0.00	0.00	435.11	792,000.00 *gal*
Application event totals		11.66	0.00	0.00	788.94	

24 - 10/23/2022: Wheat, silage, soft dough

Field name: 24

Crop: Wheat, silage, soft dough

Plant date: 10/23/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/10/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	127.01	24.38	209.22	0.00	56.00 *ton*
Application event totals		127.01	24.38	209.22	0.00	

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24 - 10/23/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/16/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	4.11	0.00	0.00	278.64	667,800.00 *gal*
Irrigation Wells	Ground water	5.07	0.00	0.00	342.65	831,600.00 *gal*
Application event totals		9.18	0.00	0.00	621.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	3.52	0.00	0.00	238.83	572,400.00 *gal*
Irrigation Wells	Ground water	4.35	0.00	0.00	293.70	712,800.00 *gal*
Application event totals		7.87	0.00	0.00	532.53	

24 - 06/01/2023: Corn, silage

Field name: 24

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/08/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	165.67	38.23	265.92	0.00	96.00 *ton*
Application event totals		165.67	38.23	265.92	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	6.46	0.00	0.00	437.86	1,049,400.00 *gal*
Irrigation Wells	Ground water	7.97	0.00	0.00	538.45	1,306,800.00 *gal*
Application event totals		14.43	0.00	0.00	976.31	

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24 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	5.68	0.00	0.00	384.79	922,200.00 *gal*
Irrigation Wells	Ground water	7.01	0.00	0.00	473.18	1,148,400.00 *gal*
Application event totals		12.68	0.00	0.00	857.97	

07/09/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	25.60	0.00	0.00	0.00	
Application event totals		25.60	0.00	0.00	0.00	

07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	6.85	0.00	0.00	464.40	1,113,000.00 *gal*
Irrigation Wells	Ground water	8.46	0.00	0.00	571.08	1,386,000.00 *gal*
Application event totals		15.31	0.00	0.00	1,035.48	

07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	8.22	0.00	0.00	557.28	1,335,600.00 *gal*
Irrigation Wells	Ground water	10.15	0.00	0.00	685.30	1,663,200.00 *gal*
Application event totals		18.37	0.00	0.00	1,242.57	

08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	5.87	0.00	0.00	398.06	954,000.00 *gal*
Irrigation Wells	Ground water	7.25	0.00	0.00	489.50	1,188,000.00 *gal*
Application event totals		13.12	0.00	0.00	887.55	

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24 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	4.70	0.00	0.00	318.45	763,200.00 *gal*
Irrigation Wells	Ground water	5.80	0.00	0.00	391.60	950,400.00 *gal*
Application event totals		10.50	0.00	0.00	710.04	

08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	5.48	0.00	0.00	371.52	890,400.00 *gal*
Irrigation Wells	Ground water	6.77	0.00	0.00	456.86	1,108,800.00 *gal*
Application event totals		12.25	0.00	0.00	828.38	

25 - 10/23/2022: Wheat, silage, soft dough

Field name: 25

Crop: Wheat, silage, soft dough

Plant date: 10/23/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/13/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	170.77	32.78	281.31	0.00	160.00 *ton*
Application event totals		170.77	32.78	281.31	0.00	

10/26/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	4.88	0.00	0.00	330.93	1,685,400.00 *gal*
Irrigation Wells	Ground water	6.03	0.00	0.00	406.95	2,098,800.00 *gal*
Application event totals		10.91	0.00	0.00	737.89	

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25 - 10/23/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	6.45	0.00	0.00	437.08	2,226,000.00 *gal*
Irrigation Wells	Ground water	7.96	0.00	0.00	537.49	2,772,000.00 *gal*
Application event totals		14.41	0.00	0.00	974.57	

25 - 06/10/2023: Corn, silage

Field name: 25

Crop: Corn, silage

Plant date: 06/10/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/08/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	110.45	25.49	177.28	0.00	136.00 *ton*
Application event totals		110.45	25.49	177.28	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	9.21	0.00	0.00	624.40	3,180,000.00 *gal*
Irrigation Wells	Ground water	11.37	0.00	0.00	767.84	3,960,000.00 *gal*
Application event totals		20.58	0.00	0.00	1,392.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	8.10	0.00	0.00	549.47	2,798,400.00 *gal*
Irrigation Wells	Ground water	10.01	0.00	0.00	675.70	3,484,800.00 *gal*
Application event totals		18.11	0.00	0.00	1,225.17	

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25 - 06/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	25.60	0.00	0.00	0.00	
Application event totals		25.60	0.00	0.00	0.00	

07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	7.64	0.00	0.00	518.25	2,639,400.00 *gal*
Irrigation Wells	Ground water	9.44	0.00	0.00	637.31	3,286,800.00 *gal*
Application event totals		17.08	0.00	0.00	1,155.56	

07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	4.97	0.00	0.00	337.18	1,717,200.00 *gal*
Irrigation Wells	Ground water	6.14	0.00	0.00	414.63	2,138,400.00 *gal*
Application event totals		11.11	0.00	0.00	751.81	

08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	4.42	0.00	0.00	299.71	1,526,400.00 *gal*
Irrigation Wells	Ground water	5.46	0.00	0.00	368.56	1,900,800.00 *gal*
Application event totals		9.88	0.00	0.00	668.28	

08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	3.68	0.00	0.00	249.76	1,272,000.00 *gal*
Irrigation Wells	Ground water	4.55	0.00	0.00	307.14	1,584,000.00 *gal*
Application event totals		8.23	0.00	0.00	556.90	

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25 - 06/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	4.70	0.00	0.00	318.45	1,621,800.00 *gal*
Irrigation Wells	Ground water	5.80	0.00	0.00	391.60	2,019,600.00 *gal*
Application event totals		10.50	0.00	0.00	710.04	

26 - 10/23/2022: Wheat, silage, soft dough

Field name: 26

Crop: Wheat, silage, soft dough

Plant date: 10/23/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/10/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	135.48	26.01	223.17	0.00	112.00 *ton*
Application event totals		135.48	26.01	223.17	0.00	

10/25/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	8.88	0.00	0.00	573.20	3,434,400.00 *gal*
Application event totals		8.88	0.00	0.00	573.20	

01/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	8.64	0.00	0.00	557.28	3,339,000.00 *gal*
Application event totals		8.64	0.00	0.00	557.28	

26 - 06/03/2023: Corn, silage

Field name: 26

Crop: Corn, silage

Plant date: 06/03/2023

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26 - 06/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/09/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	158.31	36.53	254.11	0.00	172.00 *ton*
Application event totals		158.31	36.53	254.11	0.00	

05/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	10.28	0.00	0.00	663.43	3,975,000.00 *gal*
Application event totals		10.28	0.00	0.00	663.43	

06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	11.60	0.00	0.00	748.35	4,483,800.00 *gal*
Application event totals		11.60	0.00	0.00	748.35	

07/09/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	25.60	0.00	0.00	0.00	
Application event totals		25.60	0.00	0.00	0.00	

07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	13.16	0.00	0.00	849.19	5,088,000.00 *gal*
Application event totals		13.16	0.00	0.00	849.19	

07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	11.60	0.00	0.00	748.35	4,483,800.00 *gal*
Application event totals		11.60	0.00	0.00	748.35	

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26 - 06/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	12.67	0.00	0.00	817.34	4,897,200.00 *gal*
Application event totals		12.67	0.00	0.00	817.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	12.26	0.00	0.00	790.81	4,738,200.00 *gal*
Application event totals		12.26	0.00	0.00	790.81	

27 - 10/24/2022: Wheat, silage, soft dough

Field name: 27

Crop: Wheat, silage, soft dough

Plant date: 10/24/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/11/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	124.42	23.88	204.95	0.00	120.00 *ton*
Application event totals		124.42	23.88	204.95	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/15/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	9.48	0.00	0.00	611.83	4,276,800.00 *gal*
Application event totals		9.48	0.00	0.00	611.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	13.43	0.00	0.00	866.75	6,058,800.00 *gal*
Application event totals		13.43	0.00	0.00	866.75	

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27 - 06/15/2023: Corn, silage

Field name: 27

Crop: Corn, silage

Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	170.41	39.32	273.52	0.00	216.00 *ton*
Application event totals		170.41	39.32	273.52	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	13.83	0.00	0.00	892.25	6,237,000.00 *gal*
Application event totals		13.83	0.00	0.00	892.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	13.30	0.00	0.00	858.26	5,999,400.00 *gal*
Application event totals		13.30	0.00	0.00	858.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	26.50	0.00	0.00	0.00	
Application event totals		26.50	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	11.33	0.00	0.00	730.79	5,108,400.00 *gal*
Application event totals		11.33	0.00	0.00	730.79	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	12.64	0.00	0.00	815.77	5,702,400.00 *gal*
Application event totals		12.64	0.00	0.00	815.77	

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27 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	10.67	0.00	0.00	688.31	4,811,400.00 *gal*
Application event totals		10.67	0.00	0.00	688.31	

09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	9.48	0.00	0.00	611.83	4,276,800.00 *gal*
Application event totals		9.48	0.00	0.00	611.83	

28 - 10/12/2022: Alfalfa, hay

Field name: 28

Crop: Alfalfa, hay

Plant date: 10/12/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	200.98	38.58	331.08	0.00	432.00 *ton*
Application event totals		200.98	38.58	331.08	0.00	

10/14/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	8.03	0.00	0.00	544.35	6,360,000.00 *gal*
Irrigation Wells	Ground water	9.91	0.00	0.00	669.40	7,920,000.00 *gal*
Application event totals		17.94	0.00	0.00	1,213.75	

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28 - 10/12/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	8.19	0.00	0.00	555.24	6,487,200.00 *gal*
Irrigation Wells	Ground water	10.11	0.00	0.00	682.79	8,078,400.00 *gal*
Application event totals		18.30	0.00	0.00	1,238.02	

06/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	6.86	0.00	0.00	465.42	5,437,800.00 *gal*
Irrigation Wells	Ground water	8.48	0.00	0.00	572.33	6,771,600.00 *gal*
Application event totals		15.34	0.00	0.00	1,037.75	

06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	3.41	0.00	0.00	231.35	2,703,000.00 *gal*
Irrigation Wells	Ground water	4.21	0.00	0.00	284.49	3,366,000.00 *gal*
Application event totals		7.63	0.00	0.00	515.84	

07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	4.54	0.00	0.00	307.56	3,593,400.00 *gal*
Irrigation Wells	Ground water	5.60	0.00	0.00	378.21	4,474,800.00 *gal*
Application event totals		10.14	0.00	0.00	685.77	

07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	4.42	0.00	0.00	299.39	3,498,000.00 *gal*
Irrigation Wells	Ground water	5.45	0.00	0.00	368.17	4,356,000.00 *gal*
Application event totals		9.87	0.00	0.00	667.56	

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28 - 10/12/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	4.74	0.00	0.00	321.17	3,752,400.00 *gal*
Irrigation Wells	Ground water	5.85	0.00	0.00	394.94	4,672,800.00 *gal*
Application event totals		10.59	0.00	0.00	716.11	

09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	4.22	0.00	0.00	285.78	3,339,000.00 *gal*
Irrigation Wells	Ground water	5.20	0.00	0.00	351.43	4,158,000.00 *gal*
Application event totals		9.42	0.00	0.00	637.22	

10/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	4.01	0.00	0.00	272.18	3,180,000.00 *gal*
Irrigation Wells	Ground water	4.96	0.00	0.00	334.70	3,960,000.00 *gal*
Application event totals		8.97	0.00	0.00	606.87	

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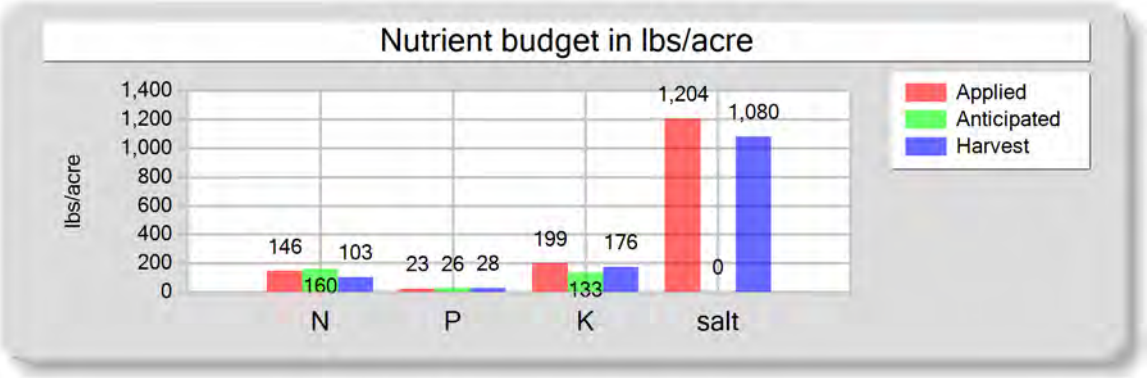
B. NUTRIENT BUDGET

23 - 10/23/2022: Wheat, silage, soft dough

Field name: 23 Crop: Wheat, silage, soft dough Plant date: 10/23/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	120.96	23.22	199.26	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	17.79	0.00	0.00	1,203.97
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	145.75	23.22	199.26	1,203.97
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	103.47	28.45	175.89	1,080.30
Nutrient balance	42.28	-5.23	23.37	123.67
Applied to removed ratio	1.41	0.82	1.13	1.11

Fresh water applied	
2,175,000.00	*gallons*
80.10	*acre-inches*
6.67	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

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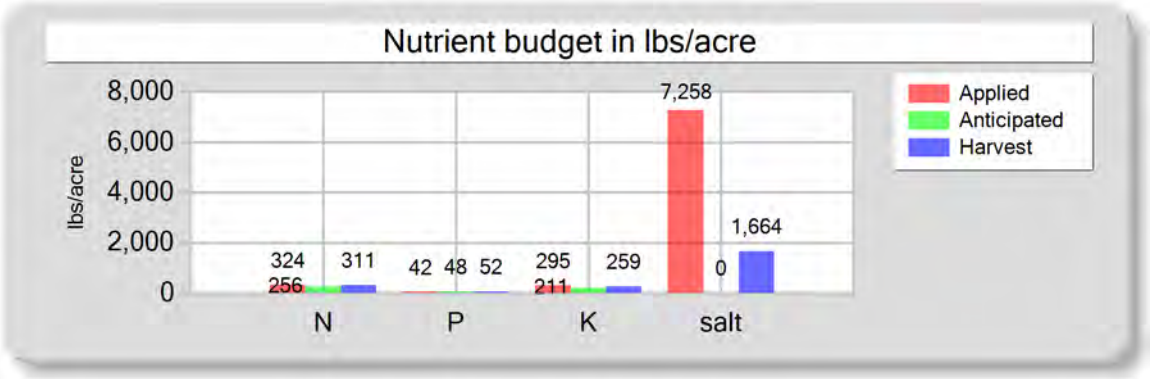
23 - 06/01/2023: Corn, silage

Field name: 23 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	25.60	0.00	0.00	0.00
Dry manure	184.08	42.48	295.47	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	107.30	0.00	0.00	7,258.21
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	323.98	42.48	295.47	7,258.21
Anticipated crop nutrient removal	256.00	48.00	211.20	0.00
Actual crop nutrient removal	311.13	51.86	259.28	1,663.68
Nutrient balance	12.85	-9.38	36.20	5,594.53
Applied to removed ratio	1.04	0.82	1.14	4.36

Fresh water applied	
13,137,600.00	*gallons*
483.81	*acre-inches*
40.32	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

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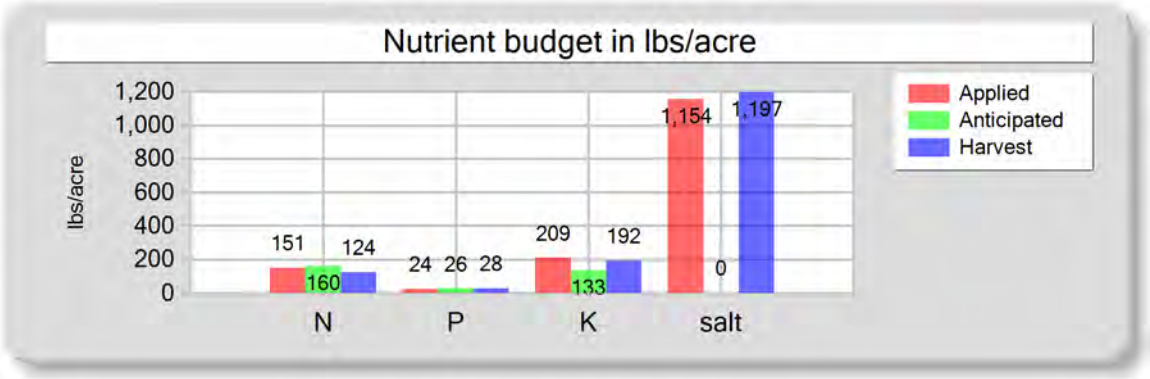
24 - 10/23/2022: Wheat, silage, soft dough

Field name: 24 Crop: Wheat, silage, soft dough Plant date: 10/23/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	127.01	24.38	209.22	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	17.06	0.00	0.00	1,153.82
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	151.07	24.38	209.22	1,153.82
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	124.10	27.92	192.36	1,196.63
Nutrient balance	26.97	-3.54	16.87	-42.81
Applied to removed ratio	1.22	0.87	1.09	0.96

Fresh water applied	
2,784,600.00	*gallons*
102.55	*acre-inches*
6.41	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

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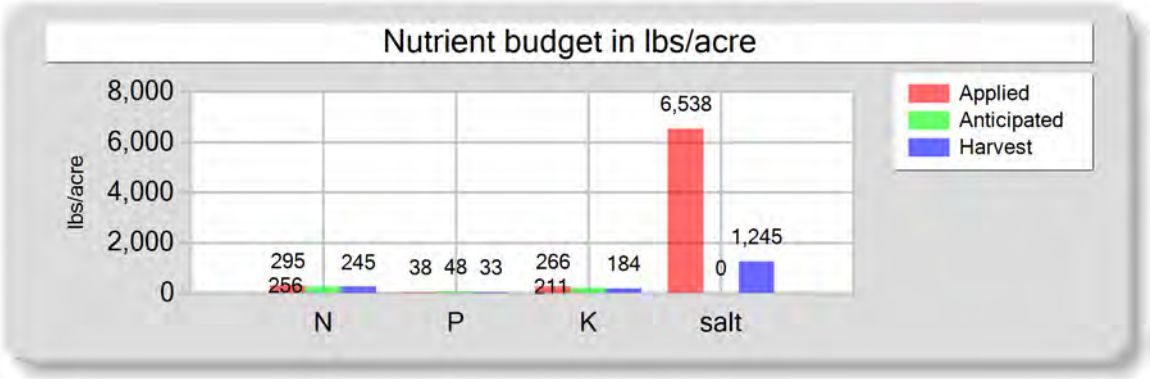
24 - 06/01/2023: Corn, silage

Field name: 24 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	25.60	0.00	0.00	0.00
Dry manure	165.67	38.23	265.92	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	96.66	0.00	0.00	6,538.31
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	294.93	38.23	265.92	6,538.31
Anticipated crop nutrient removal	256.00	48.00	211.20	0.00
Actual crop nutrient removal	245.25	32.70	183.94	1,244.89
Nutrient balance	49.68	5.53	81.99	5,293.42
Applied to removed ratio	1.20	1.17	1.45	5.25

Fresh water applied	
15,779,400.00	*gallons*
581.10	*acre-inches*
36.32	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

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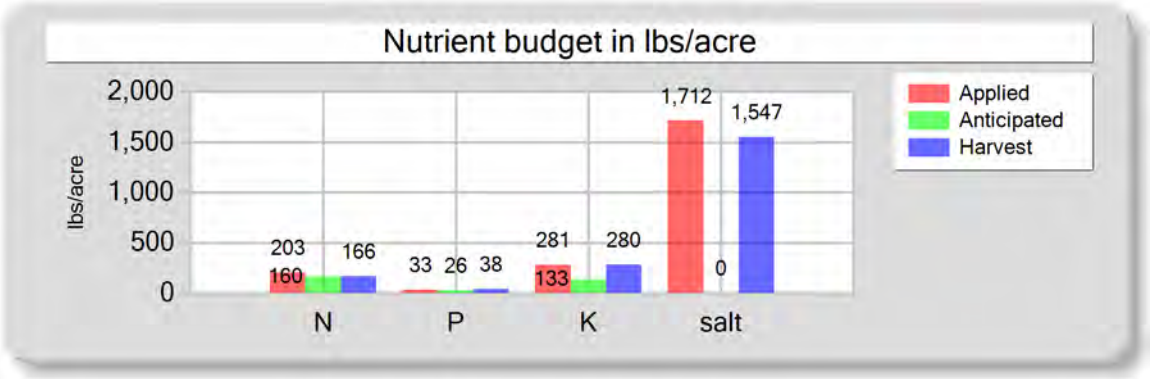
25 - 10/23/2022: Wheat, silage, soft dough

Field name: 25 Crop: Wheat, silage, soft dough Plant date: 10/23/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	170.77	32.78	281.31	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	25.32	0.00	0.00	1,712.46
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	203.08	32.78	281.31	1,712.46
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	165.51	38.20	280.10	1,547.48
Nutrient balance	37.57	-5.41	1.21	164.97
Applied to removed ratio	1.23	0.86	1.00	1.11

Fresh water applied	
8,782,200.00	*gallons*
323.42	*acre-inches*
9.51	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

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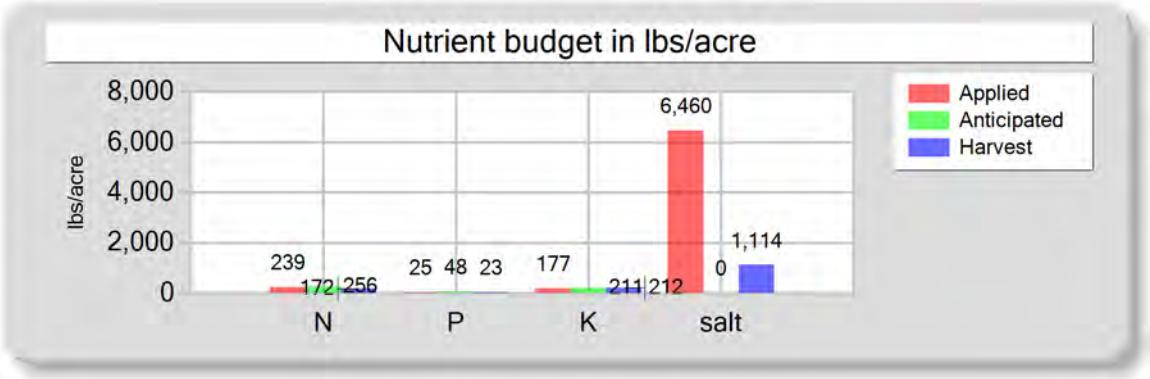
25 - 06/10/2023: Corn, silage

Field name: 25 Crop: Corn, silage Plant date: 06/10/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	25.60	0.00	0.00	0.00
Dry manure	110.45	25.49	177.28	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	95.50	0.00	0.00	6,460.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	238.55	25.49	177.28	6,460.00
Anticipated crop nutrient removal	256.00	48.00	211.20	0.00
Actual crop nutrient removal	171.69	22.59	212.36	1,113.75
Nutrient balance	66.85	2.90	-35.07	5,346.25
Applied to removed ratio	1.39	1.13	0.83	5.80

Fresh water applied	
33,129,600.00	*gallons*
1,220.05	*acre-inches*
35.88	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

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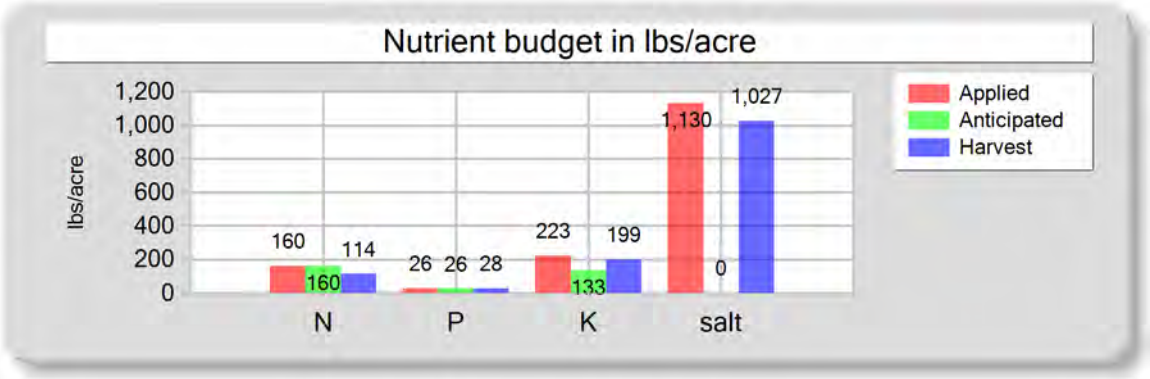
26 - 10/23/2022: Wheat, silage, soft dough

Field name: 26 Crop: Wheat, silage, soft dough Plant date: 10/23/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	135.48	26.01	223.17	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	17.52	0.00	0.00	1,130.48
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	160.00	26.01	223.17	1,130.48
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	113.71	28.43	198.99	1,026.77
Nutrient balance	46.29	-2.42	24.18	103.71
Applied to removed ratio	1.41	0.91	1.12	1.10

Fresh water applied	
6,773,400.00	*gallons*
249.44	*acre-inches*
8.31	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

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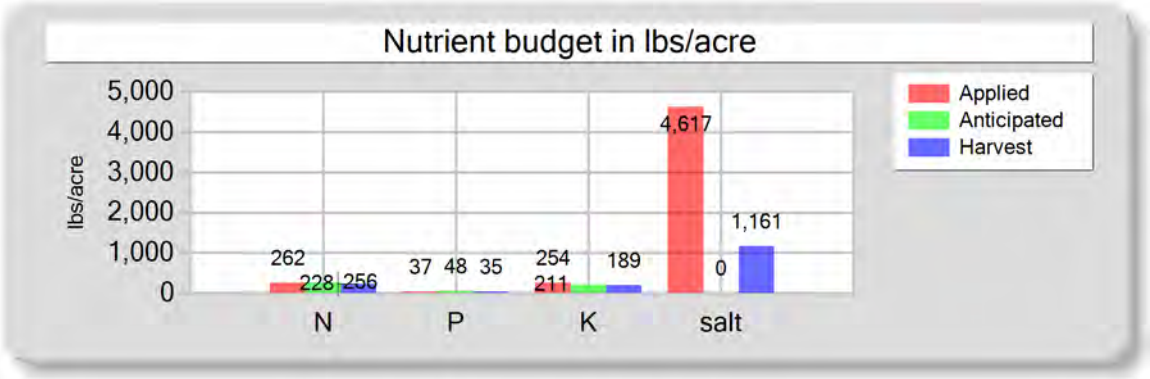
26 - 06/03/2023: Corn, silage

Field name: 26 Crop: Corn, silage Plant date: 06/03/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	25.60	0.00	0.00	0.00
Dry manure	158.31	36.53	254.11	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	71.57	0.00	0.00	4,617.46
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	262.48	36.53	254.11	4,617.46
Anticipated crop nutrient removal	256.00	48.00	211.20	0.00
Actual crop nutrient removal	228.21	35.11	188.71	1,160.89
Nutrient balance	34.27	1.42	65.39	3,456.57
Applied to removed ratio	1.15	1.04	1.35	3.98

Fresh water applied	
27,666,000.00	*gallons*
1,018.84	*acre-inches*
33.96	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

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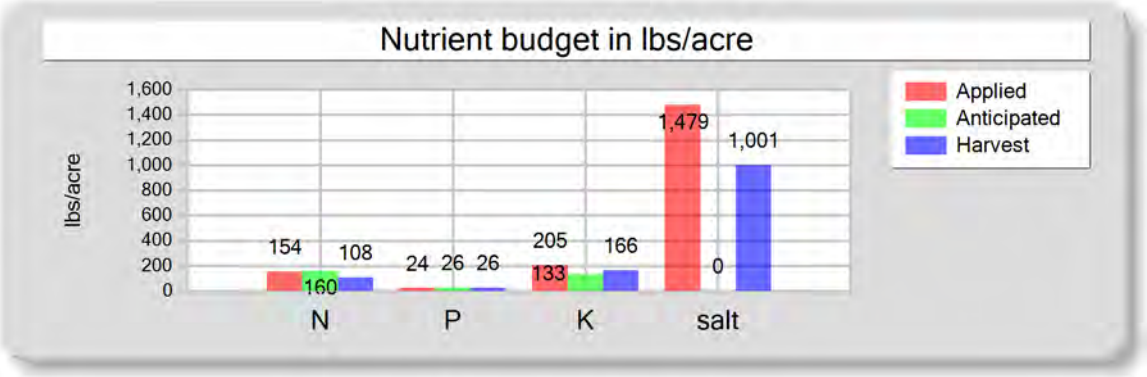
27 - 10/24/2022: Wheat, silage, soft dough

Field name: 27 Crop: Wheat, silage, soft dough Plant date: 10/24/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	124.42	23.88	204.95	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	22.92	0.00	0.00	1,478.58
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	154.33	23.88	204.95	1,478.58
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	107.89	26.31	165.78	1,001.02
Nutrient balance	46.45	-2.43	39.17	477.56
Applied to removed ratio	1.43	0.91	1.24	1.48

Fresh water applied	
10,335,600.00	*gallons*
380.63	*acre-inches*
10.88	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

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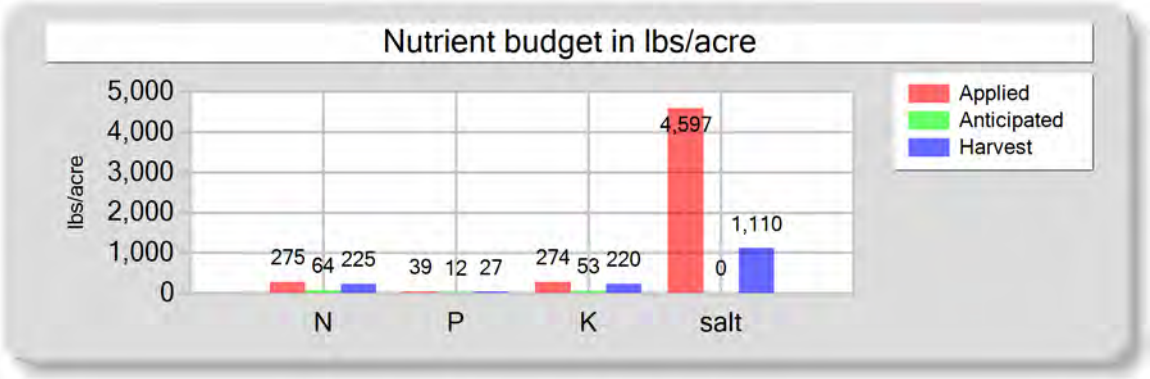
27 - 06/15/2023: Corn, silage

Field name: 27 Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	26.50	0.00	0.00	0.00
Dry manure	170.41	39.32	273.52	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	71.26	0.00	0.00	4,597.20
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	275.16	39.32	273.52	4,597.20
Anticipated crop nutrient removal	64.00	12.00	52.80	0.00
Actual crop nutrient removal	225.27	27.47	219.77	1,109.63
Nutrient balance	49.90	11.85	53.75	3,487.57
Applied to removed ratio	1.22	1.43	1.24	4.14

Fresh water applied	
32,135,400.00	*gallons*
1,183.44	*acre-inches*
33.81	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

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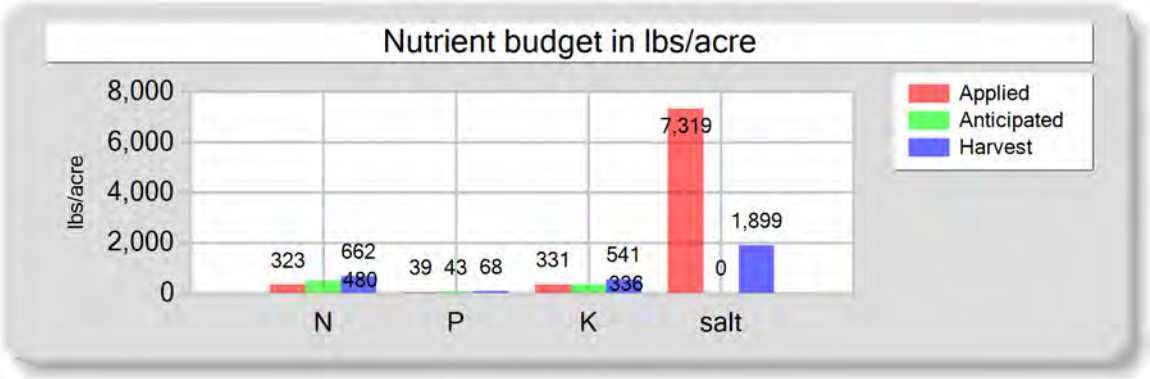
28 - 10/12/2022: Alfalfa, hay

Field name: 28 Crop: Alfalfa, hay Plant date: 10/12/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	200.98	38.58	331.08	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	108.20	0.00	0.00	7,318.90
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	323.18	38.58	331.08	7,318.90
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	661.60	67.66	541.31	1,898.80
Nutrient balance	-338.43	-29.08	-210.23	5,420.10
Applied to removed ratio	0.49	0.57	0.61	3.85

Fresh water applied	
86,108,400.00	*gallons*
3,171.08	*acre-inches*
40.65	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

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NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Solids

Sample and source description: Corral Solids
Sample date: 09/29/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 19.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,400.00	4,300.00	36,900.00							
DL	100.00	100.00	100.00							

Corral Solids

Sample and source description: Corral Solids
Sample date: 04/19/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 29.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,500.00	4,500.00	31,300.00							
DL	100.00	100.00	100.00							

Corral Solids

Sample and source description: Corral Solids
Sample date: 10/13/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 34.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,100.00	4,800.00	30,400.00	20,700.00	7,700.00	5,200.00	4,700.00	13,000.00		32.20
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	1,000.00		0.01

B. PROCESS WASTEWATER ANALYSES

No process wastewater analyses entered.

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C. FRESH WATER ANALYSES

Irrigation Wells

IW #13

Sample description: IW #13

Sample date: 08/21/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	11.80	0.00	11.60								1,160.00	800
DL	1.00	0.50	0.10								10.00	10

IW #14

Sample description: IW #14

Sample date: 08/21/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	11.70	0.00	11.60								1,170.00	790
DL	1.00	0.50	0.10								10.00	10

IW #15

Sample description: IW #15

Sample date: 08/21/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	9.30	0.00	9.10								943.00	600
DL	1.00	0.50	0.10								10.00	10

D. SOIL ANALYSES

23

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

23

Soil Sample

Sample and source description: Soil Sample

Sample date: 04/11/2024 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			26.00				
DL			2.00				

24

Soil Sample

Sample and source description: Soil Sample

Sample date: 04/11/2024 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			26.00				
DL			2.00				

25

Soil Sample

Sample and source description: Soil Sample

Sample date: 04/11/2024 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			23.00				
DL			2.00				

26

26

Soil Sample

Sample and source description: Soil Sample

Sample date: 04/11/2024 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			19.00				
DL			2.00				

27

Soil Sample

Sample and source description: Soil Sample

Sample date: 04/11/2024 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			25.00				
DL			2.00				

28

Soil Sample

Sample and source description: Soil Sample

Sample date: 04/11/2024 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			23.00				
DL			2.00				

E. PLANT TISSUE ANALYSES

23 - 10/23/2022: Wheat, silage, soft dough

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

23 - 10/23/2022: Wheat, silage, soft dough

Winter Forage

Sample and source description: Winter Forage
Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 60.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,000.00	1,100.00	6,800.00		10.60
DL	100.00	100.00	100.00		0.01

23 - 06/01/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 09/14/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 62.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,400.00	900.00	4,500.00		7.70
DL	100.00	100.00	100.00		0.01

24 - 10/23/2022: Wheat, silage, soft dough

Winter Forage

Sample and source description: Winter Forage
Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 59.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,000.00	900.00	6,200.00		9.50
DL	100.00	100.00	100.00		0.01

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

24 - 06/01/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 09/14/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 57.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,000.00	800.00	4,500.00		7.20
DL	100.00	100.00	100.00		0.01

25 - 10/23/2022: Wheat, silage, soft dough

Winter Forage

Sample and source description: Winter Forage
Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 56.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,200.00	1,200.00	8,800.00		11.10
DL	100.00	100.00	100.00		0.01

25 - 06/10/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 09/17/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 57.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,800.00	500.00	4,700.00		5.80
DL	100.00	100.00	100.00		0.01

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

26 - 10/23/2022: Wheat, silage, soft dough

Winter Forage

Sample and source description: Winter Forage
Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 65.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,000.00	1,000.00	7,000.00		10.50
DL	100.00	100.00	100.00		0.01

26 - 06/03/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 09/14/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 61.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,200.00	800.00	4,300.00		6.80
DL	100.00	100.00	100.00		0.01

27 - 10/24/2022: Wheat, silage, soft dough

Winter Forage

Sample and source description: Winter Forage
Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 65.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,100.00	1,000.00	6,300.00		10.90
DL	100.00	100.00	100.00		0.01

27 - 06/15/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/11/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 62.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,100.00	500.00	4,000.00		5.40
DL	100.00	100.00	100.00		0.01

28 - 10/12/2022: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 11/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 54.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,600.00	1,800.00	14,400.00		11.20
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

No subsurface (tile) drainage analyses entered.

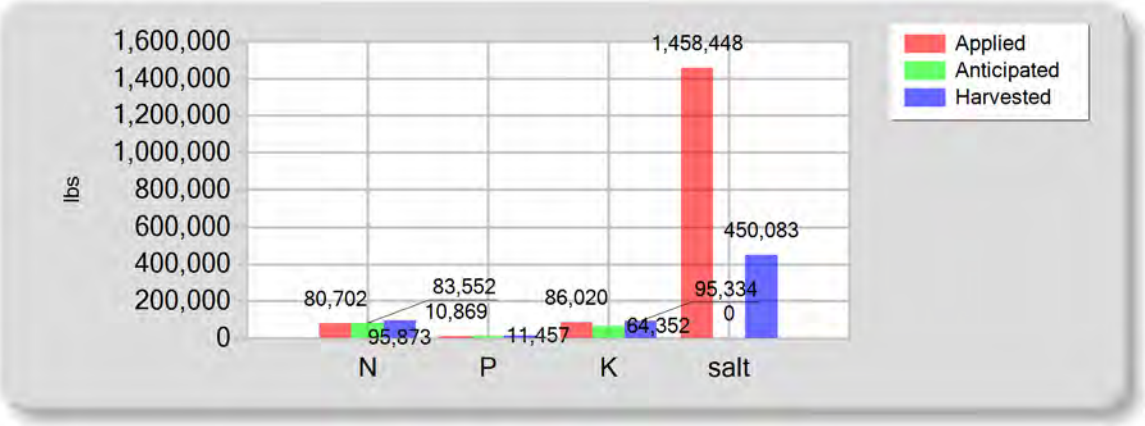
Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

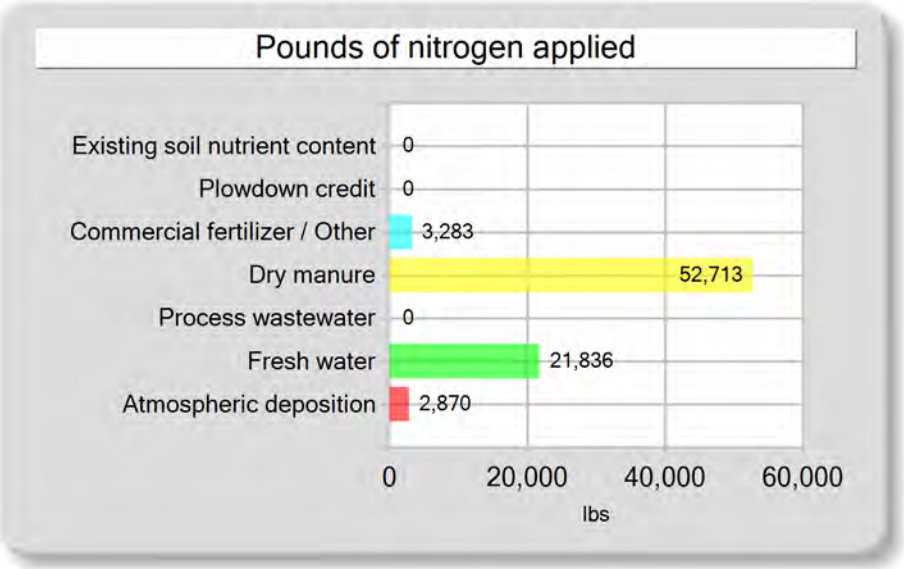
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	3,282.70	0.00	0.00	0.00
Dry manure	52,713.36	10,869.12	86,020.32	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	21,836.39	0.00	0.00	1,458,448.27
Atmospheric deposition	2,870.00	0.00	0.00	0.00
Total nutrients applied	80,702.45	10,869.12	86,020.32	1,458,448.27
Anticipated crop nutrient removal	83,552.00	11,456.80	64,352.00	0.00
Actual crop nutrient removal	95,872.64	13,066.76	95,333.72	450,082.57
Nutrient balance	-15,170.19	-2,197.64	-9,313.40	1,008,365.71
Applied to removed ratio	0.84	0.83	0.90	3.24

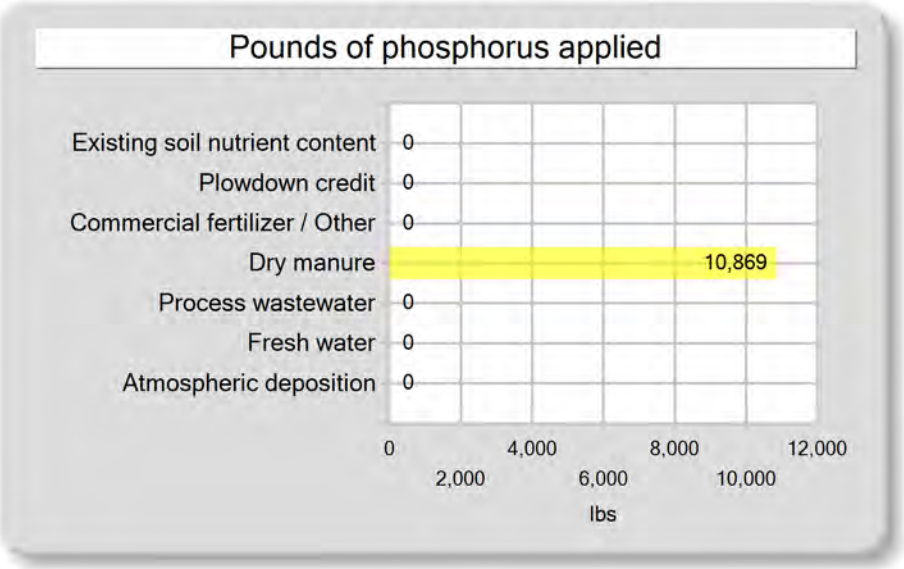
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

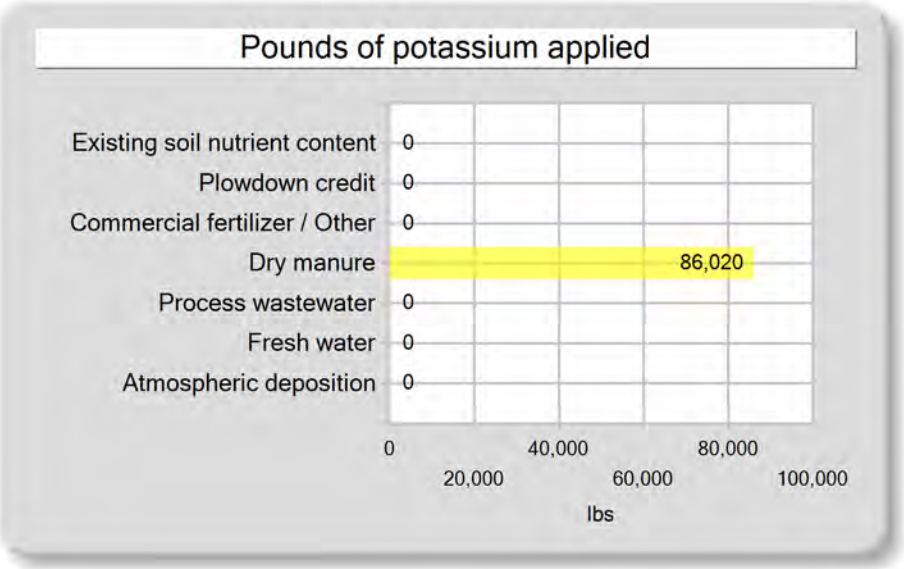


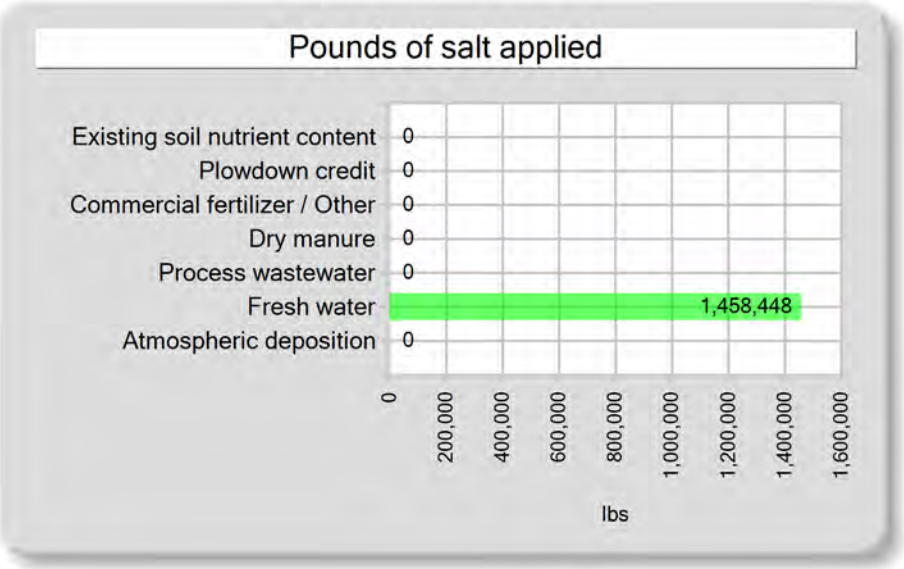
Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

ADDITIONAL NOTES

A. NOTES

Precipitation utilized during winter months to meet forage freshwater requirements. Abnormally heavy rains led to less water needing to be applied to winter crop.

Fields 23 Wheat, 26 Wheat & 27 Wheat had lower than anticipated removal rates due to lower than anticipated %N & low tonnage. The low removal rate resulted in a field ratio slightly exceeding target limits, even though application rates did not exceed expected removals.

Field 25 Corn had a lower than anticipated removal rate. This was due to lower than expected yield and a lower than expected %N. The %N was based on analysis that was derived through a certified laboratory. However, the applications to these fields matched the low removal rates and was able to meet the field ratio threshold of 1.4.

No wastewater was generated on the facility. All water runoff was evaporated throughout the year. No wastewater was utilized or exported offsite in 2023.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

DocuSigned by:
59609E2E7E0947E...

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Case Vyfhuizen	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6/20/2024	
DATE	DATE

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

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DELLAVALLE™
LABORATORY INC

5H Farms #2
5851 S Highway 59
Merced, CA 95341

Account# 00-0022208
Account Manager: Ben Nydam
Submitted By:

Received: 08/21/2023 14:30
Reported: 08/28/2023 13:28

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23H1829-01	IW #13	Drinking Water	Case V.		08/21/2023 0:00
23H1829-02	IW #14	Drinking Water	Case V.		08/21/2023 0:00
23H1829-03	IW #15	Drinking Water	Case V.		08/21/2023 0:00
23H1829-04	5H #2 D1	Drinking Water	Case V.		08/21/2023 0:00

Default Cooler Temperature on Receipt °C: 6.0
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

1910 W. McKinley Ave Suite 110 Fresno, CA 93728 559-233-6129 www.dellavallelab.com

DELLAVALLE™ LABORATORY INC

5H Farms #2
5851 S Highway 59
Merced, CA 95341

Account# 00-0022208
Account Manager: Ben Nydam
Submitted By:

Received: 08/21/2023 14:30
Reported: 08/28/2023 13:28

Sample Results

Sample: IW #13
23H1829-01 (Water)

Sampled: 8/21/2023 0:00
Sampled By: Case V.

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.16**	mmhos/cm	0.01	1		08/23/23 12:46	SM 2510 B		BEH1082
Electrical Conductivity umhos	**1160**	umhos/cm	10.0	1		08/23/23 12:46	SM 2510 B		BEH1082
Nitrate Nitrogen as NO3N	**11.6**	mg/L	0.1	1	10	08/22/23 02:47	EPA 300.0		BEH0986
pH	**8.0**	units	1.0	1		08/23/23 12:46	SM 4500-H+	H	BEH1082
Total Filterable Solids (TDS)	**800**	mg/L	10.0	1		08/24/23 12:07	SM 2540 C		BEH1056
Temperature	**25.0**	°C	0.0	1		08/23/23 12:46	SM 2510 B		BEH1082
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		08/25/23 13:23	SM 4500-NH3 C		BEH1135
Total Nitrogen	**11.8**	mg/L	1.00	1		08/25/23 13:23	SM 4500-NH3 C		BEH1135

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DELLAVALLE™ LABORATORY INC

5H Farms #2
5851 S Highway 59
Merced, CA 95341

Account# 00-0022208
Account Manager: Ben Nydam
Submitted By:

Received: 08/21/2023 14:30
Reported: 08/28/2023 13:28

Sample Results
(Continued)

Sample: IW #14
23H1829-02 (Water)

Sampled: 8/21/2023 0:00
Sampled By: Case V.

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.17**	mmhos/cm	0.01	1		08/23/23 12:48	SM 2510 B		BEH1082
Electrical Conductivity umhos	**1170**	umhos/cm	10.0	1		08/23/23 12:48	SM 2510 B		BEH1082
Nitrate Nitrogen as NO3N	**11.6**	mg/L	0.1	1	10	08/22/23 03:06	EPA 300.0		BEH0986
pH	**8.0**	units	1.0	1		08/23/23 12:48	SM 4500-H+	H	BEH1082
Total Filterable Solids (TDS)	**790**	mg/L	10.0	1		08/24/23 12:07	SM 2540 C		BEH1056
Temperature	**25.0**	°C	0.0	1		08/23/23 12:48	SM 2510 B		BEH1082
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		08/25/23 13:24	SM 4500-NH3 C		BEH1135
Total Nitrogen	**11.7**	mg/L	1.00	1		08/25/23 13:24	SM 4500-NH3 C		BEH1135

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DELLAVALLE™ LABORATORY INC

5H Farms #2
5851 S Highway 59
Merced, CA 95341

Account# 00-0022208
Account Manager: Ben Nydam
Submitted By:

Received: 08/21/2023 14:30
Reported: 08/28/2023 13:28

Sample Results
(Continued)

Sample: IW #15
23H1829-03 (Water)

Sampled: 8/21/2023 0:00
Sampled By: Case V.

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.94**	mmhos/cm	0.01	1		08/23/23 13:03	SM 2510 B		BEH1083
Electrical Conductivity umhos	**943**	umhos/cm	10.0	1		08/23/23 13:03	SM 2510 B		BEH1083
Nitrate Nitrogen as NO3N	**9.1**	mg/L	0.1	1	10	08/22/23 03:26	EPA 300.0		BEH0986
pH	**8.0**	units	1.0	1		08/23/23 13:03	SM 4500-H+	H	BEH1083
Total Filterable Solids (TDS)	**600**	mg/L	10.0	1		08/24/23 12:07	SM 2540 C		BEH1056
Temperature	**25.0**	°C	0.0	1		08/23/23 13:03	SM 2510 B		BEH1083
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		08/25/23 13:26	SM 4500-NH3 C		BEH1135
Total Nitrogen	**9.33**	mg/L	1.00	1		08/25/23 13:26	SM 4500-NH3 C		BEH1135

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DELLAVALLE™ LABORATORY INC

5H Farms #2
5851 S Highway 59
Merced, CA 95341

Account# 00-0022208
Account Manager: Ben Nydam
Submitted By:

Received: 08/21/2023 14:30
Reported: 08/28/2023 13:28

Sample Results
(Continued)

Sample: 5H #2 D1
23H1829-04 (Water)

Sampled: 8/21/2023 0:00
Sampled By: Case V.

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.51**	mmhos/cm	0.01	1		08/23/23 13:05	SM 2510 B		BEH1083
Electrical Conductivity umhos	**1510**	umhos/cm	10.0	1		08/23/23 13:05	SM 2510 B		BEH1083
Ammonia (as N)	*	mg/L	0.00	1		08/21/23 00:00	Field		BEH1301
Nitrate Nitrogen as NO3N	**19.4**	mg/L	0.1	1	10	08/22/23 03:46	EPA 300.0		BEH0986
pH	**7.9**	units	1.0	1		08/23/23 13:05	SM 4500-H+	H	BEH1083
Temperature	**25.0**	°C	0.0	1		08/23/23 13:05	SM 2510 B		BEH1083

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DELLAVALLE™ LABORATORY INC

5H Farms #2
5851 S Highway 59
Merced, CA 95341

Account# 00-0022208
Account Manager: Ben Nydam
Submitted By:

Received: 08/21/2023 14:30
Reported: 08/28/2023 13:28

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH0986										
Blank (BEH0986-BLK1)					Prepared & Analyzed: 8/21/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEH0986-BLK2)					Prepared & Analyzed: 8/21/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEH0986-BLK3)					Prepared & Analyzed: 8/22/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEH0986-BS1)					Prepared & Analyzed: 8/21/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		103	90-110		
LCS (BEH0986-BS2)					Prepared & Analyzed: 8/22/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000		103	90-110		
Duplicate (BEH0986-DUP1)		**Source: 23H1823-06**			Prepared & Analyzed: 8/21/2023					
Nitrate Nitrogen as NO3N	5.0		0.1	mg/L		5.0			0.101	10
Duplicate (BEH0986-DUP2)		**Source: 23H1823-13**			Prepared & Analyzed: 8/22/2023					
Nitrate Nitrogen as NO3N	6.4		0.1	mg/L		6.4			0.0466	10
Matrix Spike (BEH0986-MS1)		**Source: 23H1823-06**			Prepared & Analyzed: 8/21/2023					
Nitrate Nitrogen as NO3N	10.1		0.1	mg/L	5.000	5.0	102	90-110		
Matrix Spike (BEH0986-MS2)		**Source: 23H1823-13**			Prepared & Analyzed: 8/22/2023					
Nitrate Nitrogen as NO3N	11.6		0.1	mg/L	5.000	6.4	104	90-110		
Reference (BEH0986-SRM1)					Prepared & Analyzed: 8/21/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		
Reference (BEH0986-SRM2)					Prepared & Analyzed: 8/22/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEH0986-SRM3)					Prepared & Analyzed: 8/22/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		

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DELLAVALLE™ LABORATORY INC

5H Farms #2
5851 S Highway 59
Merced, CA 95341

Account# 00-0022208
Account Manager: Ben Nydam
Submitted By:

Received: 08/21/2023 14:30
Reported: 08/28/2023 13:28

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH1056										
Blank (BEH1056-BLK1)					Prepared: 8/22/2023	Analyzed: 8/24/2023				
Total Filterable Solids (TDS)	ND		10.0	mg/L						
LCS (BEH1056-BS1)					Prepared: 8/22/2023	Analyzed: 8/24/2023				
Total Filterable Solids (TDS)	31.2		10.0	mg/L	2000		1.56	0-200		
Duplicate (BEH1056-DUP1)			**Source: 23H1762-01**		Prepared: 8/22/2023	Analyzed: 8/24/2023				
Total Filterable Solids (TDS)	200		10.0	mg/L		198			1.26	10
Duplicate (BEH1056-DUP2)			**Source: 23H1829-01**		Prepared: 8/22/2023	Analyzed: 8/24/2023				
Total Filterable Solids (TDS)	810		10.0	mg/L		800			1.24	10
Reference (BEH1056-SRM1)					Prepared: 8/22/2023	Analyzed: 8/24/2023				
Total Filterable Solids (TDS)	320			mg/L	325.0		98.5	90-110		
Reference (BEH1056-SRM2)					Prepared: 8/22/2023	Analyzed: 8/24/2023				
Total Filterable Solids (TDS)	487			mg/L	495.0		98.3	90-110		

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5851 S Highway 59
Merced, CA 95341

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Submitted By:

Received: 08/21/2023 14:30
Reported: 08/28/2023 13:28

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH1082										
Blank (BEH1082-BLK1)					Prepared: 8/22/2023	Analyzed: 8/23/2023				
pH	5.3		1.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	°C						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Blank (BEH1082-BLK2)					Prepared: 8/22/2023	Analyzed: 8/23/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	6.0		1.0	units						
Temperature	25.0		0.0	°C						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Blank (BEH1082-BLK3)					Prepared: 8/22/2023	Analyzed: 8/23/2023				
pH	6.1		1.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Duplicate (BEH1082-DUP1)		**Source: 23H1823-14**			Prepared: 8/22/2023	Analyzed: 8/23/2023				
Electrical Conductivity	0.79		0.01	mmhos/cm		0.80			1.28	10
pH	8.0		1.0	units		7.9			0.126	10
Electrical Conductivity umhos	792		10.0	umhos/cm		802			1.28	10
Duplicate (BEH1082-DUP2)		**Source: 23H1829-01**			Prepared: 8/22/2023	Analyzed: 8/23/2023				
pH	8.0		1.0	units		8.0			0.00	10
Electrical Conductivity	1.17		0.01	mmhos/cm		1.16			0.249	10
Electrical Conductivity umhos	1170		10.0	umhos/cm		1160			0.249	10
Reference (BEH1082-SRM1)					Prepared: 8/22/2023	Analyzed: 8/23/2023				
Electrical Conductivity	522			umhos/cm	538.0		97.0	90-110		
Reference (BEH1082-SRM2)					Prepared: 8/22/2023	Analyzed: 8/23/2023				
pH	5.8			units	5.820		100	28178-101.71		
Reference (BEH1082-SRM3)					Prepared: 8/22/2023	Analyzed: 8/23/2023				
Electrical Conductivity	980			umhos/cm	1000		98.0	90-110		
Electrical Conductivity umhos	980			umhos/cm	1000		98.0	90-110		
Reference (BEH1082-SRM4)					Prepared: 8/22/2023	Analyzed: 8/23/2023				
Electrical Conductivity	980			umhos/cm	1000		98.0	90-110		
Electrical Conductivity umhos	980			umhos/cm	1000		98.0	90-110		
Reference (BEH1082-SRM5)					Prepared: 8/22/2023	Analyzed: 8/23/2023				
Electrical Conductivity	983			umhos/cm	1000		98.3	90-110		

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Submitted By:

Received: 08/21/2023 14:30
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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH1082 (Continued)										
Reference (BEH1082-SRM5)					Prepared: 8/22/2023	Analyzed: 8/23/2023				
Electrical Conductivity umhos	983			umhos/cm	1000		98.3	90-110		
Reference (BEH1082-SRM6)					Prepared: 8/22/2023	Analyzed: 8/23/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEH1082-SRM7)					Prepared: 8/22/2023	Analyzed: 8/23/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEH1082-SRM8)					Prepared: 8/22/2023	Analyzed: 8/23/2023				
pH	4.0			units	4.000		100	97.5-102.5		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH1083										
Blank (BEH1083-BLK1)					Prepared: 8/22/2023	Analyzed: 8/23/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	5.5		1.0	units						
Temperature	25.0		0.0	°C						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Blank (BEH1083-BLK2)					Prepared: 8/22/2023	Analyzed: 8/23/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	6.0		1.0	units						
Temperature	25.0		0.0	°C						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Blank (BEH1083-BLK3)					Prepared: 8/22/2023	Analyzed: 8/23/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	6.0		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Duplicate (BEH1083-DUP1)		**Source: 23H1832-05**			Prepared: 8/22/2023	Analyzed: 8/23/2023				
pH	8.6		1.0	units		8.6			0.00	10
Electrical Conductivity	1.39		0.01	mmhos/cm		1.39			0.122	10
Electrical Conductivity umhos	1390		10.0	umhos/cm		1390			0.122	10
Duplicate (BEH1083-DUP2)		**Source: 23H1832-09**			Prepared: 8/22/2023	Analyzed: 8/23/2023				
pH	8.2		1.0	units		8.2			0.366	10
Electrical Conductivity	3.79		0.01	mmhos/cm		4.03			6.16	10
Electrical Conductivity umhos	3790		10.0	umhos/cm		4030			6.16	10
Reference (BEH1083-SRM1)					Prepared: 8/22/2023	Analyzed: 8/23/2023				
Electrical Conductivity	525			umhos/cm	538.0		97.6	90-110		
Reference (BEH1083-SRM2)					Prepared: 8/22/2023	Analyzed: 8/23/2023				
pH	5.8			units	5.820		101	28178-101.71		
Reference (BEH1083-SRM3)					Prepared: 8/22/2023	Analyzed: 8/23/2023				
Electrical Conductivity	986			umhos/cm	1000		98.6	90-110		
Electrical Conductivity umhos	986			umhos/cm	1000		98.6	90-110		
Reference (BEH1083-SRM4)					Prepared: 8/22/2023	Analyzed: 8/23/2023				
Electrical Conductivity	982			umhos/cm	1000		98.2	90-110		
Electrical Conductivity umhos	982			umhos/cm	1000		98.2	90-110		
Reference (BEH1083-SRM5)					Prepared: 8/22/2023	Analyzed: 8/23/2023				
Electrical Conductivity	968			umhos/cm	1000		96.8	90-110		

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5H Farms #2
5851 S Highway 59
Merced, CA 95341

Account# 00-0022208
Account Manager: Ben Nydam
Submitted By:

Received: 08/21/2023 14:30
Reported: 08/28/2023 13:28

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH1083 (Continued)										
Reference (BEH1083-SRM5)					Prepared: 8/22/2023	Analyzed: 8/23/2023				
Electrical Conductivity umhos	968			umhos/cm	1000		96.8	90-110		
Reference (BEH1083-SRM6)					Prepared: 8/22/2023	Analyzed: 8/23/2023				
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEH1083-SRM7)					Prepared: 8/22/2023	Analyzed: 8/23/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEH1083-SRM8)					Prepared: 8/22/2023	Analyzed: 8/23/2023				
pH	4.0			units	4.000		100	97.5-102.5		

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DELLAVALLE™ LABORATORY INC

5H Farms #2
5851 S Highway 59
Merced, CA 95341

Account# 00-0022208
Account Manager: Ben Nydam
Submitted By:

Received: 08/21/2023 14:30
Reported: 08/28/2023 13:28

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH1135										
Blank (BEH1135-BLK1)					Prepared: 8/23/2023	Analyzed: 8/25/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.00	mg/L						
Total Nitrogen	ND		1.00	mg/L						
Blank (BEH1135-BLK2)					Prepared: 8/23/2023	Analyzed: 8/25/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.00	mg/L						
Total Nitrogen	ND		1.00	mg/L						
LCS (BEH1135-BS1)					Prepared: 8/23/2023	Analyzed: 8/25/2023				
Kjeldahl Nitrogen (TKN), Total	5.92		1.00	mg/L	5.709		104	90-110		
LCS (BEH1135-BS2)					Prepared: 8/23/2023	Analyzed: 8/25/2023				
Kjeldahl Nitrogen (TKN), Total	5.93		1.00	mg/L	5.709		104	90-110		
Duplicate (BEH1135-DUP1)		**Source: 23H1826-09**			Prepared: 8/23/2023	Analyzed: 8/25/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.40	mg/L		ND				10
Duplicate (BEH1135-DUP2)		**Source: 23H1829-03**			Prepared: 8/23/2023	Analyzed: 8/25/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.40	mg/L		ND				10
Matrix Spike (BEH1135-MS1)		**Source: 23H1826-09**			Prepared: 8/23/2023	Analyzed: 8/25/2023				
Kjeldahl Nitrogen (TKN), Total	8.30		1.40	mg/L	7.992	ND	104	90-110		
Matrix Spike (BEH1135-MS2)		**Source: 23H1829-03**			Prepared: 8/23/2023	Analyzed: 8/25/2023				
Kjeldahl Nitrogen (TKN), Total	7.73		1.40	mg/L	7.992	ND	96.7	90-110		
Reference (BEH1135-SRM1)					Prepared: 8/23/2023	Analyzed: 8/25/2023				
Kjeldahl Nitrogen (TKN), Total	23.3			mg/L	23.80		97.9	90-110		

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23H1829

WATER WORK REQUEST

DELLAVALLE LABORATORY, INC.
1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

No. Samples: 3 4 No of Bottles: 7

Purchase Order No

Bill To: Acct # 22208 Cons # 8

Results Need By

Water Type: [X] Drinking Water [ ] Wastewater
[ ] Ag Water [X] Groundwater [ ] Monitoring Well

Name: 5H Farms #2
Address: 5851 S Highway 59
City: Merced **State:** CA **Zip:** 95341
Telephone: **Fax:**
Cell/Email: c5huizen07@gmail.com
COPY TO: krhodes@fragservices.com
ariordan@fragservices.com
REQUESTED BY:
PROJECT:
CROP:

[X] Copy of Chain [X] QA/QC Documents
Sampled By: Case V.

Other:

Analysis and Bottles Required: (*Please indicate Analysis*)

(X) DWW1: EC, NO_3-N NH4-N Field Test
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DWW2: DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
(X) DCW1: EC, NO_3-N, TKN, TN, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW1: EC, NO_3-N, NH_4-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW2: DPW1 Plus Ca, Mg, Na, HCO_3, CO_3, SO_4, Cl
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) Other

	Description of Samples		Date Sampled	Time Sampled	Rec'd Temp °C	Field NH_4-N
1	1030 11.6 IW #13	DCW 1	8/21		6.0	
2	11.6 IW #14	DCW 1	↓		7.1	
3	9.1 IW #15	DCW 1	↓		6.8	
4	9.4 5H#2 D1	DWW 1	↓		6.1	
5						
6						
7						
8				Lara 8/22/23		
9	08/21/23 14:30 23H1829			12:22 - per Alex, ok & thx		
10			EDD			

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	Kes	PR Ag	8/21 0800	8/21 0900
Second	JCF	DLV	8-21-23 12:38	8-21-23 21:00
Third	MM	DLI	8/21/23 14:30	
Fourth				

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through *cal* under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:
Sampling hrs
Consulting

Shipping
$ In
$ Out

Amt Paid	Rec By	Check #	Date

Signature

Sample received in cooler with ice (coolant)
[ ] Yes [ ] No

IR Thermometer SN: 200560723
Correction Factor: 0°C
Calibration Due: 9/26/2023
Location: Laboratory

X:\1-CLIENT CUSTOM FIELDSHEETS\FIELDSHEETS\Dairy Fieldsheets\[5H Farms #2 Water.xls]Dairy Water

DocuSign Envelope ID: C05E14F9-C239-49D2-A2DF-3C506A8DD910

08/21/23 14:30 23H1829

Shipping Information: Shipped In ☐ Picked-Up ☐ Walk In ☐ DLI Sampler ☑ Other ☐ _____

☐ Samples refridgerated before pick up ☐ Picked up samples placed in Ice chest

Container: Ice Chest ☑ Box ☐ None ☐ **Refrigerant:** Wet Ice ☑ Blue Ice ☐ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☐ Preserved Upon Receipt at Laboratory

	Type of Container(s) Received	Sample Number 1	2	3	4	5	6	7	8	9	10
	Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	* pH Value										
	250 mL H_2SO_4 (Yellow) Plastic	1	1	1							
	* pH Value	<2	<2	<2							
	500 mL unpreserved (White) Plastic										
	1 L unpreserved (White) Plastic	1	1	1	1						
	1 L unpreserved (BOD) (Purple) Plastic										
Special	500mL unpreserved (White) Glass										
	PO4-P Kit										
	Other:										
	Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	250 mL H_2SO_4 (Yellow) Plastic										
	500 mL HNO_3 (Red)										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
	1 L HNO_3 (Red)										
VOA Vials	40 mL VOA, $Na_2S_2O_3$ + MCAA (EPA531)										
	40 mL VOA, $Na_2S_2O_3$ (EPA547)										
	40mL AG VOA unpreserved (White) (Set of 3)										
	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
	40mL VOA, H_3PO_4 (Set of 3)										
	40 mL VOA, HCl (Blue) (Set of 3)										
	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
	250 mL AG H_2SO_4 (Yellow)										
	250 mL AG $Na_2S_2O_3$ (Green)										
	250 mL AG $Na_2S_2O_3$ + MCAA										
	500 mL glass unpreserved (White)										
	500 mL AG HCl (Blue)										
	1 L AG unpreserved (White)										
	1 L AG H_2SO_4 (Yellow)										
	1 L AG $Na_2S_2O_3$ (Green)										
	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
	Cyanide - 500 mL NaOH										
	Asbestos - 1L P wrapped in foil (Set of 2)										
	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
	Chlorite/Bromate - 250 mL AG with EDA										
	HAA5 - 250mL AG Ammonium Chlorite										
	DO KIT										
	Other:										
	Other:										

pH Strips
Lot 10BDH4501
Exp. Jan. 2025

SEND OUT